

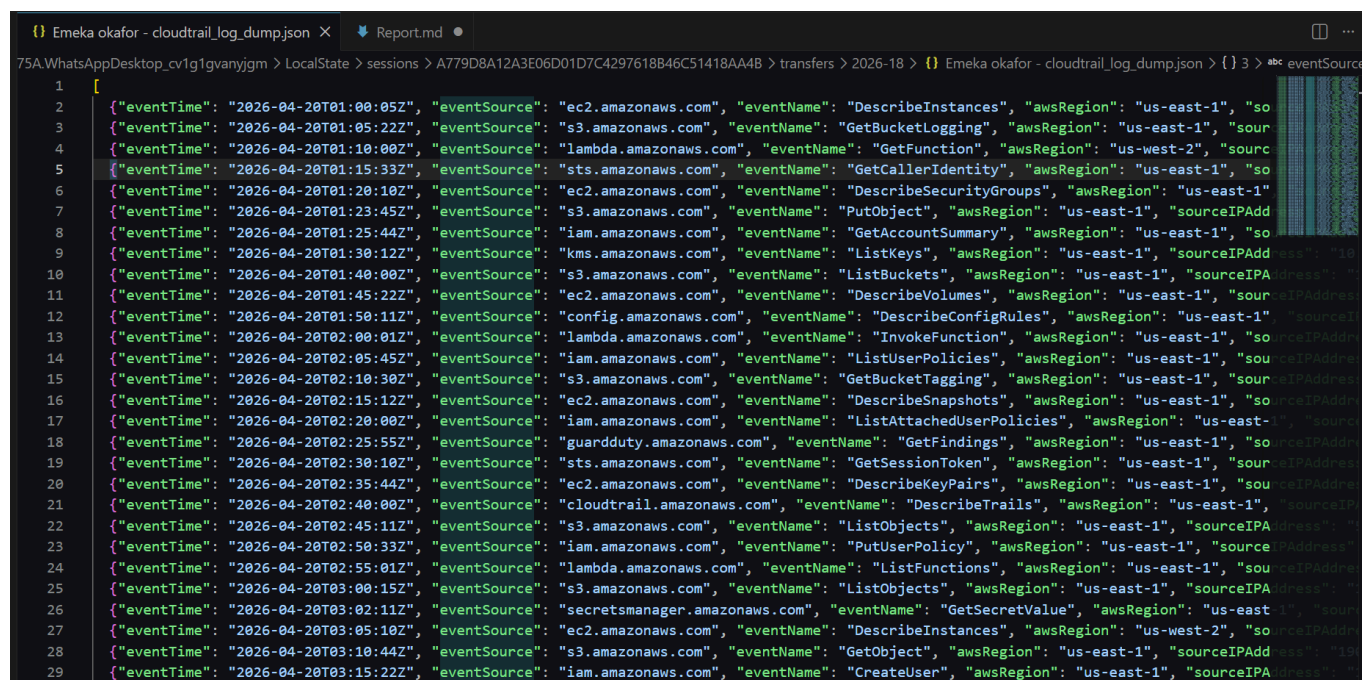
Cloud Forensic Investigation Report – Operation Overcast

1. Introduction

This report presents findings from an investigation into a suspected AWS cloud compromise using CloudTrail logs. The objective is to identify unauthorized access, attacker behavior, and data exfiltration.

2. Methodology

The analysis was conducted using VS Code for log inspection and CyberChef for decoding Base64-encoded data.



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1 [{"eventTime": "2026-04-20T01:00:05Z", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeInstances", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
2 [{"eventTime": "2026-04-20T01:05:22Z", "eventSource": "s3.amazonaws.com", "eventName": "GetBucketLogging", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
3 [{"eventTime": "2026-04-20T01:10:00Z", "eventSource": "lambda.amazonaws.com", "eventName": "GetFunction", "awsRegion": "us-west-2", "sourceIP": "193.50.135.100"}]
4 [{"eventTime": "2026-04-20T01:15:33Z", "eventSource": "sts.amazonaws.com", "eventName": "GetCallerIdentity", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
5 [{"eventTime": "2026-04-20T01:20:10Z", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeSecurityGroups", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
6 [{"eventTime": "2026-04-20T01:23:45Z", "eventSource": "s3.amazonaws.com", "eventName": "PutObject", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
7 [{"eventTime": "2026-04-20T01:25:44Z", "eventSource": "iam.amazonaws.com", "eventName": "GetAccountSummary", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
8 [{"eventTime": "2026-04-20T01:30:12Z", "eventSource": "kms.amazonaws.com", "eventName": "ListKeys", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
9 [{"eventTime": "2026-04-20T01:40:00Z", "eventSource": "s3.amazonaws.com", "eventName": "ListBuckets", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
10 [{"eventTime": "2026-04-20T01:45:22Z", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeVolumes", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
11 [{"eventTime": "2026-04-20T02:10:30Z", "eventSource": "config.amazonaws.com", "eventName": "DescribeConfigRules", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
12 [{"eventTime": "2026-04-20T02:00:01Z", "eventSource": "lambda.amazonaws.com", "eventName": "InvokeFunction", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
13 [{"eventTime": "2026-04-20T02:05:45Z", "eventSource": "iam.amazonaws.com", "eventName": "ListUserPolicies", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
14 [{"eventTime": "2026-04-20T02:15:12Z", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeSnapshots", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
15 [{"eventTime": "2026-04-20T02:25:55Z", "eventSource": "iam.amazonaws.com", "eventName": "ListAttachedUserPolicies", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
16 [{"eventTime": "2026-04-20T02:25:55Z", "eventSource": "guardduty.amazonaws.com", "eventName": "GetFindings", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
17 [{"eventTime": "2026-04-20T02:30:10Z", "eventSource": "sts.amazonaws.com", "eventName": "GetSessionToken", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
18 [{"eventTime": "2026-04-20T02:35:44Z", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeKeyPairs", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
19 [{"eventTime": "2026-04-20T02:40:00Z", "eventSource": "cloudtrail.amazonaws.com", "eventName": "DescribeTrails", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
20 [{"eventTime": "2026-04-20T02:45:11Z", "eventSource": "s3.amazonaws.com", "eventName": "ListObjects", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
21 [{"eventTime": "2026-04-20T02:50:33Z", "eventSource": "iam.amazonaws.com", "eventName": "PutUserPolicy", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
22 [{"eventTime": "2026-04-20T02:55:01Z", "eventSource": "lambda.amazonaws.com", "eventName": "ListFunctions", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
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24 [{"eventTime": "2026-04-20T03:02:11Z", "eventSource": "secretsmanager.amazonaws.com", "eventName": "GetSecretValue", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
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26 [{"eventTime": "2026-04-20T03:10:44Z", "eventSource": "s3.amazonaws.com", "eventName": "GetObject", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
27 [{"eventTime": "2026-04-20T03:15:22Z", "eventSource": "iam.amazonaws.com", "eventName": "CreateUser", "awsRegion": "us-east-1", "sourceIP": "193.50.135.100"}]
```

3. Credential Discovery

The compromised access key identified during analysis is:

AKIA_DEV_PROD_9982

This key (highlighted) was observed being used from an external IP address, indicating unauthorized access.

```
Emeka okafor - cloudtrail_log_dump.json • Report.md •
> Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > A779D8A12A3E06D01D7C4297618B46C51418AA4B > transfers > 2026-18 > {} Emeka okafor - cloudtrail_log_dump.json > ...

> accessKeyId 1 of 20

1
2 s-east-1", "sourceIPAddress": "172.31.45.12", "userAgent": "aws-internal/3", "userIdentity": {"type": "AssumedRole", "principalId": "AROA:S
3 east-1", "sourceIPAddress": "52.94.23.11", "userAgent": "S3Console/0.1", "userIdentity": {"type": "IAMUser", "userName": "admin_backup"}},
4 est-2", "sourceIPAddress": "10.0.1.55", "userAgent": "Boto3/1.26.1", "userIdentity": {"type": "AssumedRole", "principalId": "AROA:LambdaExe
5 s-east-1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"type": "IAMUser", "accessKeyId": "AKIA_DEV_P
6 ": "us-east-1", "sourceIPAddress": "172.31.45.12", "userAgent": "aws-internal/3"},
7 , "sourceIPAddress": "10.0.1.55", "userAgent": "DevOps-Deployer/1.0", "requestParameters": {"bucketName": "app-configs", "key": "prod_conf
8 s-east-1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"accessKeyId": "AKIA_DEV_PROD_9982"}},
9 , "sourceIPAddress": "10.0.5.10", "userAgent": "Boto3/1.26.1"},
10 1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"accessKeyId": "AKIA_DEV_PROD_9982"}},
11 east-1", "sourceIPAddress": "172.31.45.12", "userAgent": "aws-internal/3"},
12 ": "us-east-1", "sourceIPAddress": "internal", "userAgent": "config.amazonaws.com"},
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15 east-1", "sourceIPAddress": "54.23.11.9", "userAgent": "S3Console/0.1"},
16 s-east-1", "sourceIPAddress": "172.31.45.12", "userAgent": "aws-internal/3"},
17 on": "us-east-1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"accessKeyId": "AKIA_DEV_PROD_9982"}, "requestParameters":
18 s-east-1", "sourceIPAddress": "internal", "userAgent": "guardduty.amazonaws.com"},
19 east-1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"accessKeyId": "AKIA_DEV_PROD_9982"}},
20 -east-1", "sourceIPAddress": "172.31.45.12", "userAgent": "aws-internal/3"},
21 : "us-east-1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"accessKeyId": "AKIA_DEV_PROD_9982"}},
22 1", "sourceIPAddress": "54.23.11.9", "userAgent": "S3Console/0.1", "requestParameters": {"bucketName": "public-assets"}},
23 st-1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"accessKeyId": "AKIA_DEV_PROD_9982"}, "requestParameters":
24 -east-1", "sourceIPAddress": "10.0.1.55"},
25 1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"accessKeyId": "AKIA_DEV_PROD_9982"}, "requestParameters": {"bu
26 ion": "us-east-1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-cli/2.0.30", "userIdentity": {"accessKeyId": "AKIA_DEV_PROD_9982"}, "response
27 s-west-2", "sourceIPAddress": "172.31.45.12", "userAgent": "aws-internal/3"},
28
```

4. IP Analysis

The initial attacker IP address identified is: 190.45.112.3

This IP (highlighted) initiated access using the compromised access key, marking the beginning of the attack.

```
?75A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > A779D8A12A3E06D01D7C4297618B46C51418AA4B > transfers > 2026-18 > {} Emeka okafor - cloudtrail_log_dump.json > {} 3 > aws events

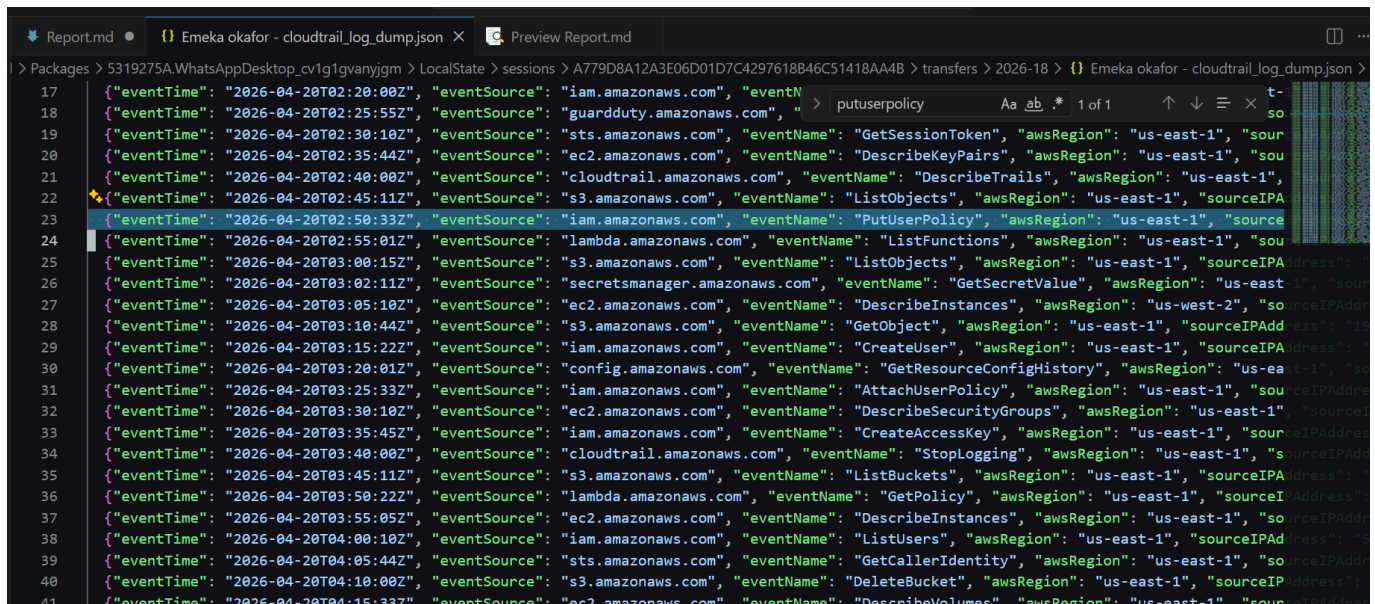
> 190.45.112.3 1 of 17

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2 ", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeInstances", "awsRegion": "us-east-1", "sourceIPAddress": "172.31.45.12", "user
3 ", "eventSource": "s3.amazonaws.com", "eventName": "GetBucketLogging", "awsRegion": "us-east-1", "sourceIPAddress": "52.94.23.11", "userAge
4 ", "eventSource": "lambda.amazonaws.com", "eventName": "GetFunction", "awsRegion": "us-west-2", "sourceIPAddress": "10.0.1.55", "userAgent":
5 ", "eventSource": "sts.amazonaws.com", "eventName": "GetCallerIdentity", "awsRegion": "us-east-1", "sourceIPAddress": "190.45.112.3", "user
6 ", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeSecurityGroups", "awsRegion": "us-east-1", "sourceIPAddress": "172.31.45.12",
7 ", "eventSource": "s3.amazonaws.com", "eventName": "PutObject", "awsRegion": "us-east-1", "sourceIPAddress": "10.0.1.55", "userAgent": "DevOps-Dep
8 ", "eventSource": "iam.amazonaws.com", "eventName": "GetAccountSummary", "awsRegion": "us-east-1", "sourceIPAddress": "190.45.112.3", "userAgent":
9 ", "eventSource": "kms.amazonaws.com", "eventName": "ListKeys", "awsRegion": "us-east-1", "sourceIPAddress": "10.0.5.10", "userAgent": "Bot
10 ", "eventSource": "s3.amazonaws.com", "eventName": "ListBuckets", "awsRegion": "us-east-1", "sourceIPAddress": "190.45.112.3", "userAgent": "aws-
11 ", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeVolumes", "awsRegion": "us-east-1", "sourceIPAddress": "172.31.45.12", "userAgent": "
12 ", "eventSource": "config.amazonaws.com", "eventName": "DescribeConfigRules", "awsRegion": "us-east-1", "sourceIPAddress": "internal", "userAgent":
13 ", "eventSource": "lambda.amazonaws.com", "eventName": "InvokeFunction", "awsRegion": "us-east-1", "sourceIPAddress": "internal", "userAgent": "aw
14 ", "eventSource": "iam.amazonaws.com", "eventName": "ListUserPolicies", "awsRegion": "us-east-1", "sourceIPAddress": "190.45.112.3", "userAgent":
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16 ", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeSnapshots", "awsRegion": "us-east-1", "sourceIPAddress": "172.31.45.12", "userAgent":
17 ", "eventSource": "iam.amazonaws.com", "eventName": "ListAttachedUserPolicies", "awsRegion": "us-east-1", "sourceIPAddress": "190.45.112.3", "user
18 ", "eventSource": "guardduty.amazonaws.com", "eventName": "GetFindings", "awsRegion": "us-east-1", "sourceIPAddress": "internal", "userAgent": "gu
19 ", "eventSource": "sts.amazonaws.com", "eventName": "GetSessionToken", "awsRegion": "us-east-1", "sourceIPAddress": "190.45.112.3", "userAgent":
20 ", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeKeyPairs", "awsRegion": "us-east-1", "sourceIPAddress": "172.31.45.12", "userAgent":
21 ", "eventSource": "cloudtrail.amazonaws.com", "eventName": "DescribeTrails", "awsRegion": "us-east-1", "sourceIPAddress": "190.45.112.3", "userAge
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```

5. Privilege Escalation

The attacker escalated privileges using the PutUserPolicy API call, granting full administrative access. This action was observed at 2026-04-20T02:50:33Z from the IP address 190.45.112.3. The applied policy granted full administrative privileges, as shown where "Action": "" and "Resource": "" were specified

screenshot 1(highlighted)



```
17 {\"eventTime\": \"2026-04-20T02:20:00Z\", \"eventSource\": \"iam.amazonaws.com\", \"eventName\": \"PutUserPolicy\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
18 {\"eventTime\": \"2026-04-20T02:25:55Z\", \"eventSource\": \"guardduty.amazonaws.com\", \"eventName\": \"PutUserPolicy\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
19 {\"eventTime\": \"2026-04-20T02:30:10Z\", \"eventSource\": \"sts.amazonaws.com\", \"eventName\": \"GetSessionToken\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
20 {\"eventTime\": \"2026-04-20T02:35:44Z\", \"eventSource\": \"ec2.amazonaws.com\", \"eventName\": \"DescribeKeyPairs\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
21 {\"eventTime\": \"2026-04-20T02:40:00Z\", \"eventSource\": \"cloudtrail.amazonaws.com\", \"eventName\": \"DescribeTrails\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
22 {\"eventTime\": \"2026-04-20T02:45:11Z\", \"eventSource\": \"s3.amazonaws.com\", \"eventName\": \"ListObjects\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
23 {\"eventTime\": \"2026-04-20T02:50:33Z\", \"eventSource\": \"iam.amazonaws.com\", \"eventName\": \"PutUserPolicy\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
24 {\"eventTime\": \"2026-04-20T02:55:01Z\", \"eventSource\": \"lambda.amazonaws.com\", \"eventName\": \"ListFunctions\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
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28 {\"eventTime\": \"2026-04-20T03:10:44Z\", \"eventSource\": \"s3.amazonaws.com\", \"eventName\": \"GetObject\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
29 {\"eventTime\": \"2026-04-20T03:15:22Z\", \"eventSource\": \"iam.amazonaws.com\", \"eventName\": \"CreateUser\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
30 {\"eventTime\": \"2026-04-20T03:20:01Z\", \"eventSource\": \"config.amazonaws.com\", \"eventName\": \"GetResourceConfigHistory\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
31 {\"eventTime\": \"2026-04-20T03:25:33Z\", \"eventSource\": \"iam.amazonaws.com\", \"eventName\": \"AttachUserPolicy\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
32 {\"eventTime\": \"2026-04-20T03:30:10Z\", \"eventSource\": \"ec2.amazonaws.com\", \"eventName\": \"DescribeSecurityGroups\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
33 {\"eventTime\": \"2026-04-20T03:35:45Z\", \"eventSource\": \"iam.amazonaws.com\", \"eventName\": \"CreateAccessKey\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
34 {\"eventTime\": \"2026-04-20T03:40:00Z\", \"eventSource\": \"cloudtrail.amazonaws.com\", \"eventName\": \"StopLogging\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
35 {\"eventTime\": \"2026-04-20T03:45:11Z\", \"eventSource\": \"s3.amazonaws.com\", \"eventName\": \"ListBuckets\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
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37 {\"eventTime\": \"2026-04-20T03:55:05Z\", \"eventSource\": \"ec2.amazonaws.com\", \"eventName\": \"DescribeInstances\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
38 {\"eventTime\": \"2026-04-20T04:00:10Z\", \"eventSource\": \"iam.amazonaws.com\", \"eventName\": \"ListUsers\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
39 {\"eventTime\": \"2026-04-20T04:05:44Z\", \"eventSource\": \"sts.amazonaws.com\", \"eventName\": \"GetCallerIdentity\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
40 {\"eventTime\": \"2026-04-20T04:10:00Z\", \"eventSource\": \"s3.amazonaws.com\", \"eventName\": \"DeleteBucket\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
41 {\"eventTime\": \"2026-04-20T04:15:33Z\", \"eventSource\": \"ec2.amazonaws.com\", \"eventName\": \"DescribeVolumes\", \"awsRegion\": \"us-east-1\", \"sourceIP\"...
```

screenshot 2 (highlighted)



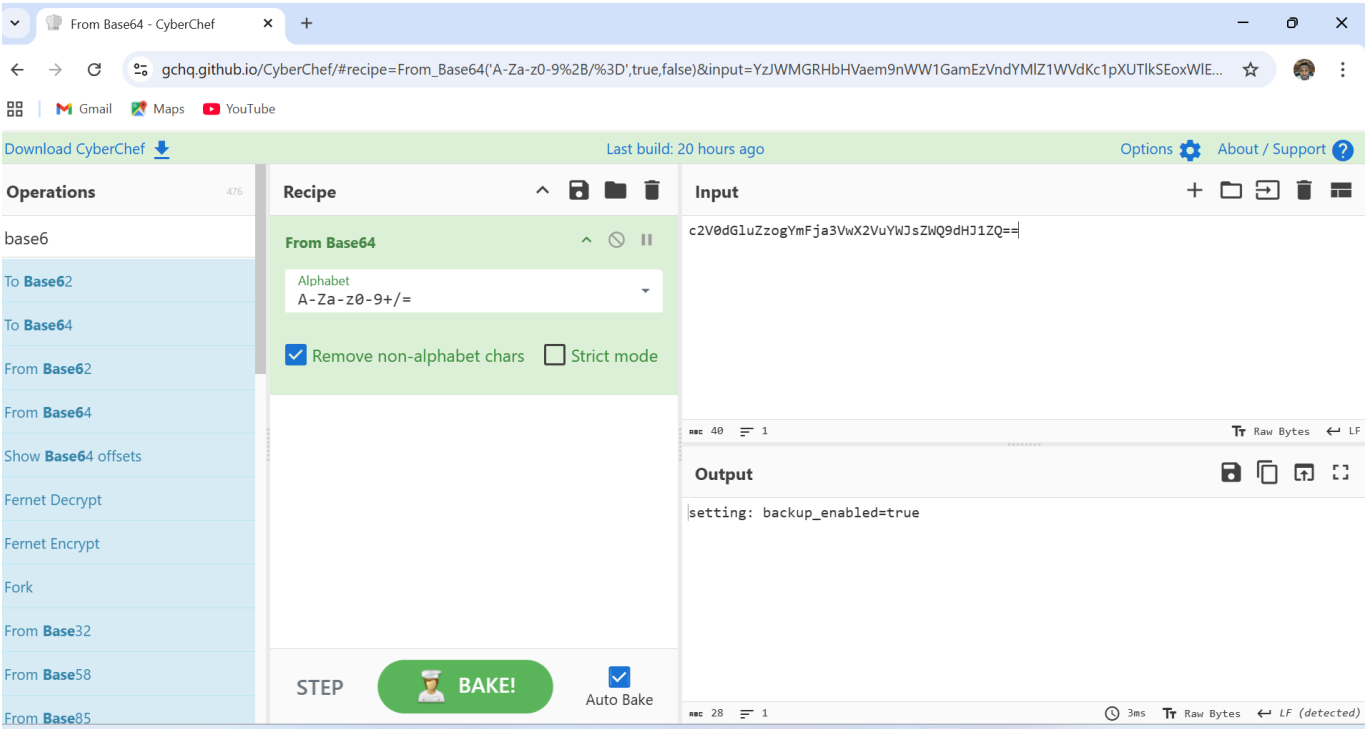
```
17 {\"dev_user_musa\"},
18
19
20
21
22
23 {\"dev_user_musa\", \"policyName\": \"FullAdmin\", \"policyDocument\": \"...\"}
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25 {\"op-secret-financials-2026\"},
26 {\"secretString\": \"ZGJfcGFzc2ogUEBzc3cwcmlRfQ2xvdWRfMjAyNiE=\"},
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28 {\"-secret-financials-2026\", \"key\": \"ceo_personal_taxes_2025.pdf\"},
29 {\"port_service_backup\"},
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6. Obfuscated Data Analysis

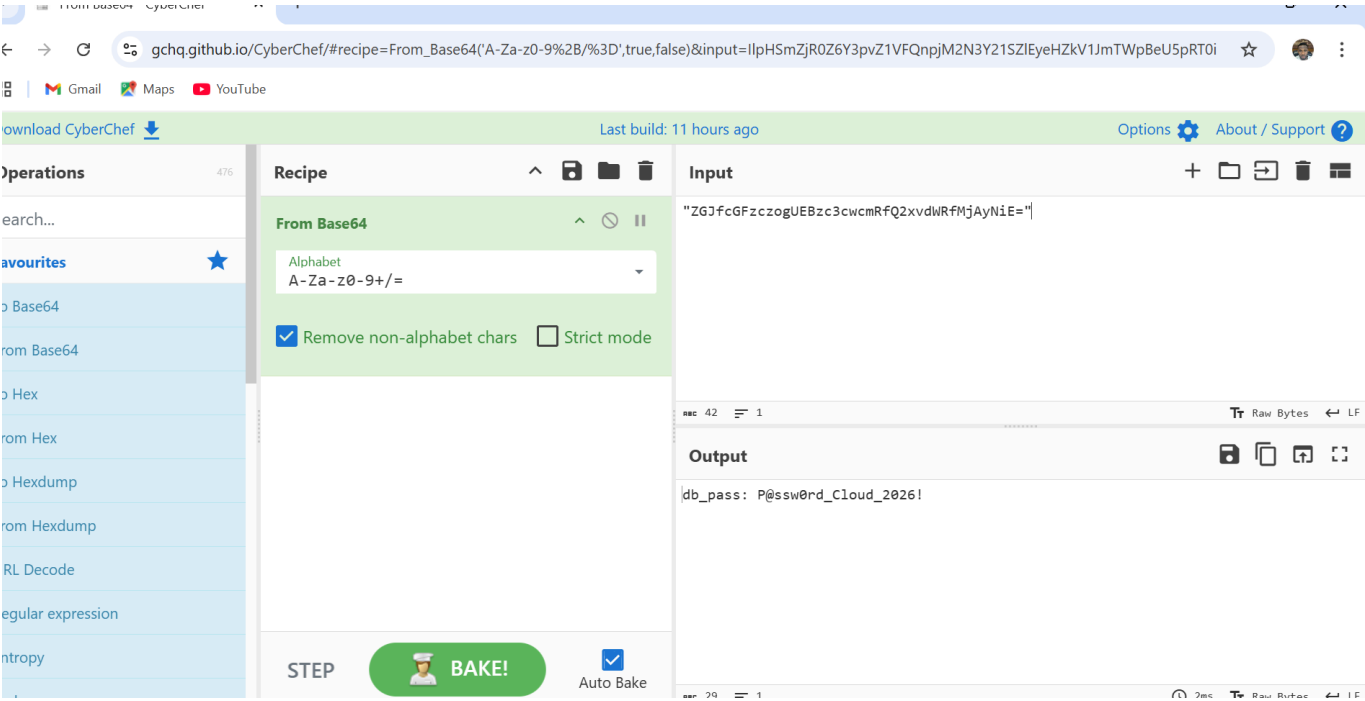
Two Base64-encoded strings were identified and decoded using CyberChef.

One contained normal configuration data, while the other exposed sensitive credentials, indicating a security breach.

screenshot 1: normal configuration data



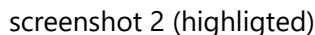
screenshot 2: this contains sensitive database credentials and it signals a data breach



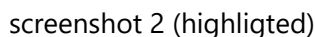
7. Data Exfiltration

The attacker downloaded a sensitive file from an S3 bucket, confirming data exfiltration activity.

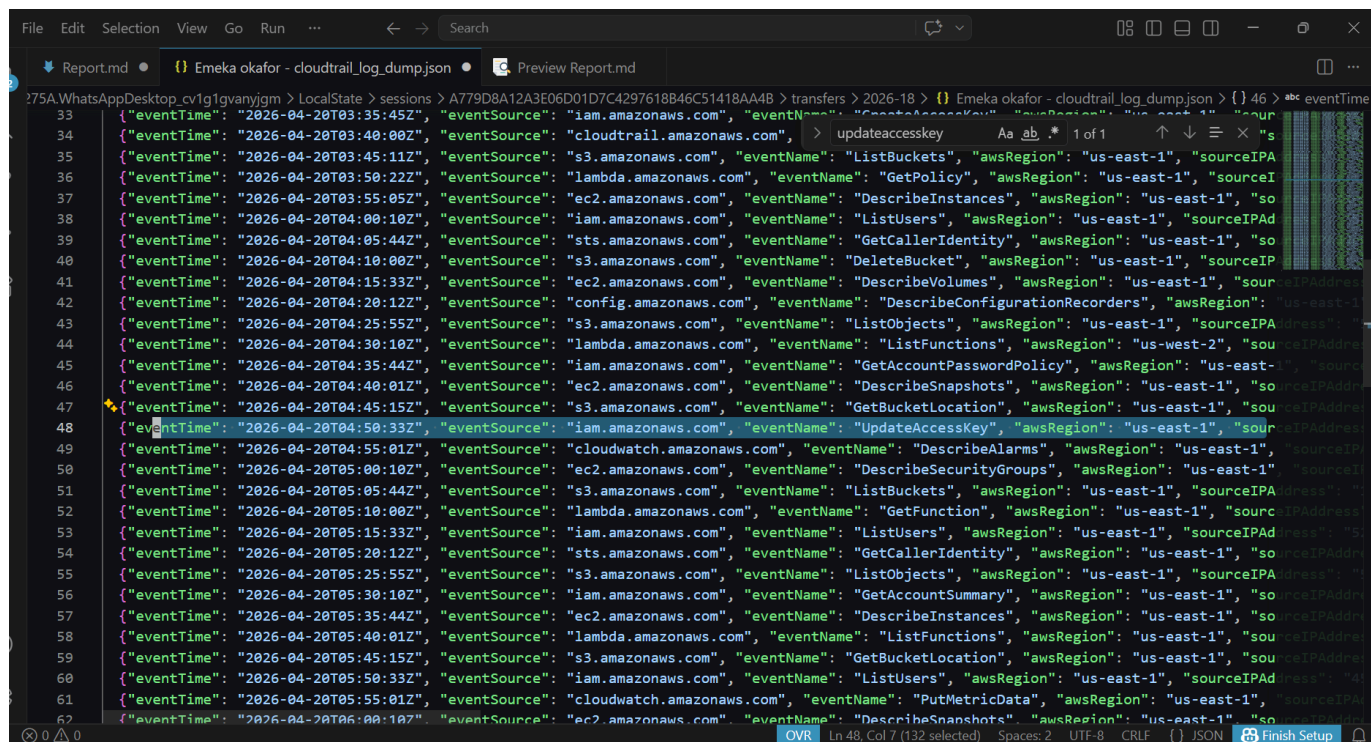
screenshot 1 (highlighted)



screenshot 1 (highlighted)

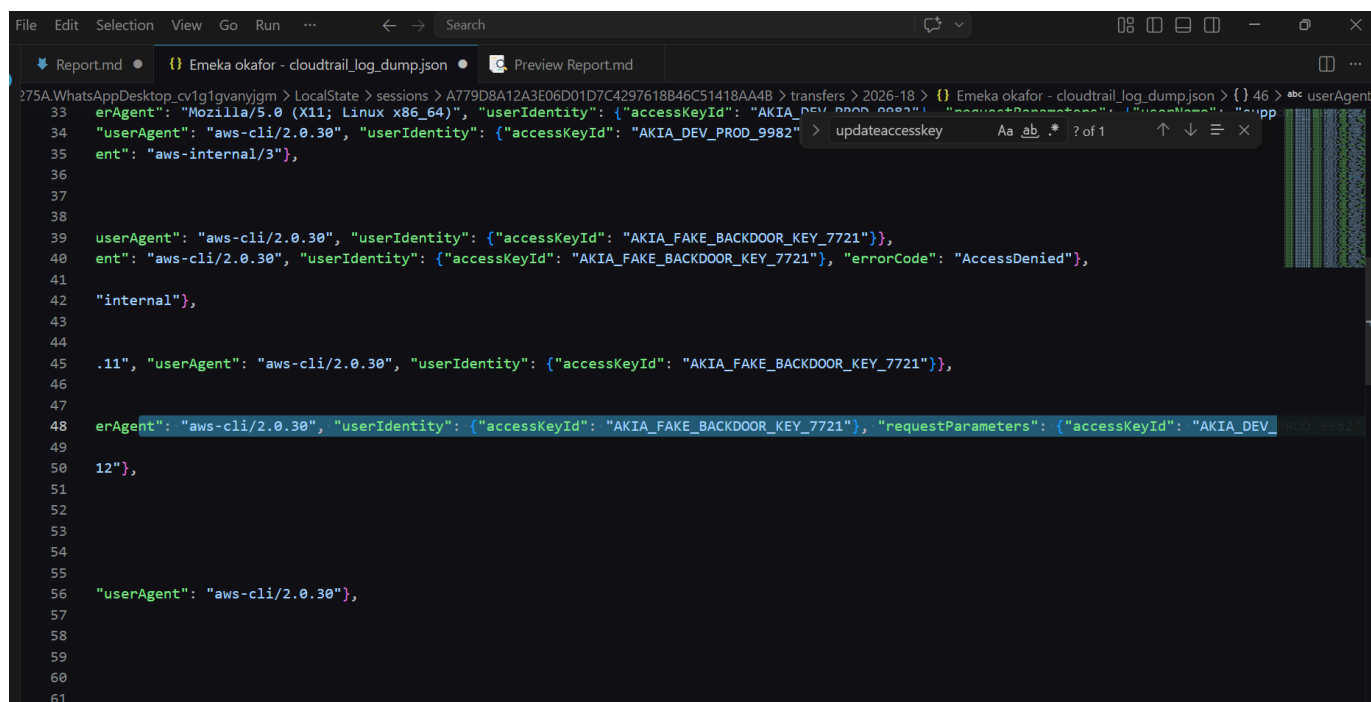


screenshot 1 (highlighted)



```
File Edit Selection View Go Run ... Search
Report.md • Emeka okafor - cloudtrail_log_dump.json • Preview Report.md
275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > A779D8A12A3E06D01D7C4297618B46C51418AA4B > transfers > 2026-18 > {} Emeka okafor - cloudtrail_log_dump.json > {} 46 > abc eventTime
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35 {"eventTime": "2026-04-20T03:45:11Z", "eventSource": "s3.amazonaws.com", "eventName": "ListBuckets", "awsRegion": "us-east-1", "sourceIP": "10.0.0.1"}
36 {"eventTime": "2026-04-20T03:50:22Z", "eventSource": "lambda.amazonaws.com", "eventName": "GetPolicy", "awsRegion": "us-east-1", "sourceIP": "10.0.0.1"}
37 {"eventTime": "2026-04-20T03:55:05Z", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeInstances", "awsRegion": "us-east-1", "sourceIP": "10.0.0.1"}
38 {"eventTime": "2026-04-20T04:00:10Z", "eventSource": "iam.amazonaws.com", "eventName": "ListUsers", "awsRegion": "us-east-1", "sourceIP": "10.0.0.1"}
39 {"eventTime": "2026-04-20T04:05:44Z", "eventSource": "sts.amazonaws.com", "eventName": "GetCallerIdentity", "awsRegion": "us-east-1", "sourceIP": "10.0.0.1"}
40 {"eventTime": "2026-04-20T04:10:00Z", "eventSource": "s3.amazonaws.com", "eventName": "DeleteBucket", "awsRegion": "us-east-1", "sourceIP": "10.0.0.1"}
41 {"eventTime": "2026-04-20T04:15:33Z", "eventSource": "ec2.amazonaws.com", "eventName": "DescribeVolumes", "awsRegion": "us-east-1", "sourceIP": "10.0.0.1"}
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OVR Ln 48, Col 7 (132 selected) Spaces: 2 UTF-8 CRLF {} JSON Finish Setup
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screenshot 2 (highlighted)



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OVR Ln 48, Col 7 (132 selected) Spaces: 2 UTF-8 CRLF {} JSON Finish Setup
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11. Conclusion

The investigation revealed that the AWS environment was compromised through the use of a stolen access key. The attacker conducted reconnaissance, escalated privileges to gain full administrative access, and successfully exfiltrated sensitive data from an S3 bucket.

Additionally, the attacker established persistence by creating a backdoor IAM user and attempted to evade detection by disabling the original access key.

This incident highlights the importance of monitoring cloud activity and securing access credentials.

12. Recommendations

To prevent similar incidents in the future, the following measures are recommended:

- Enforce Multi-Factor Authentication (MFA) for all IAM users
- Rotate and manage access keys regularly
- Apply the principle of least privilege for IAM roles and users
- Enable continuous monitoring using AWS CloudTrail and GuardDuty
- Avoid storing sensitive credentials in plain or encoded formats
- Implement alerting for suspicious activities such as privilege escalation and unusual IP access